Assignment : 7

Q.1. Accept 10 array elements from user. Sort the array in ascending order. Display sorted array.
Create class SortArray. And use functions for

1. taking input from user
2. sorting array
3. display that sorted array

Q.2. WAP to find Factorial of a Number. Create class FindFactorial and functions to

1. Accept a number from user.
2. Calculate factorial using loop.
3. Display result.

Q.3. WAP for Decimal to Binary Conversion. Create a class DecimalToBinary and functions to

1. Convert decimal number to binary.
2. Display binary equivalent

Q.4. WAP to print Hollow Square Pattern. Take integer value n from user. If n = 4, then it should print the pattern as follows :

```
* * * *
*     *
*     *
* * * *
```

Q.5. Create a menu driven program for Student Record System. Use map to store rollNo and name.

Menu options:

1. Add Record
2. Display Records
3. Save to File
4. Exit

Q.6. Create a menu driven program for Shopping Cart System. Create class Item with data members itemId, itemName, price. Use **vector** to store items.

Menu options:

1. Add Item
2. Display Items
3. Calculate Total Bill
4. Exit

Throw exception if price is invalid or negative value